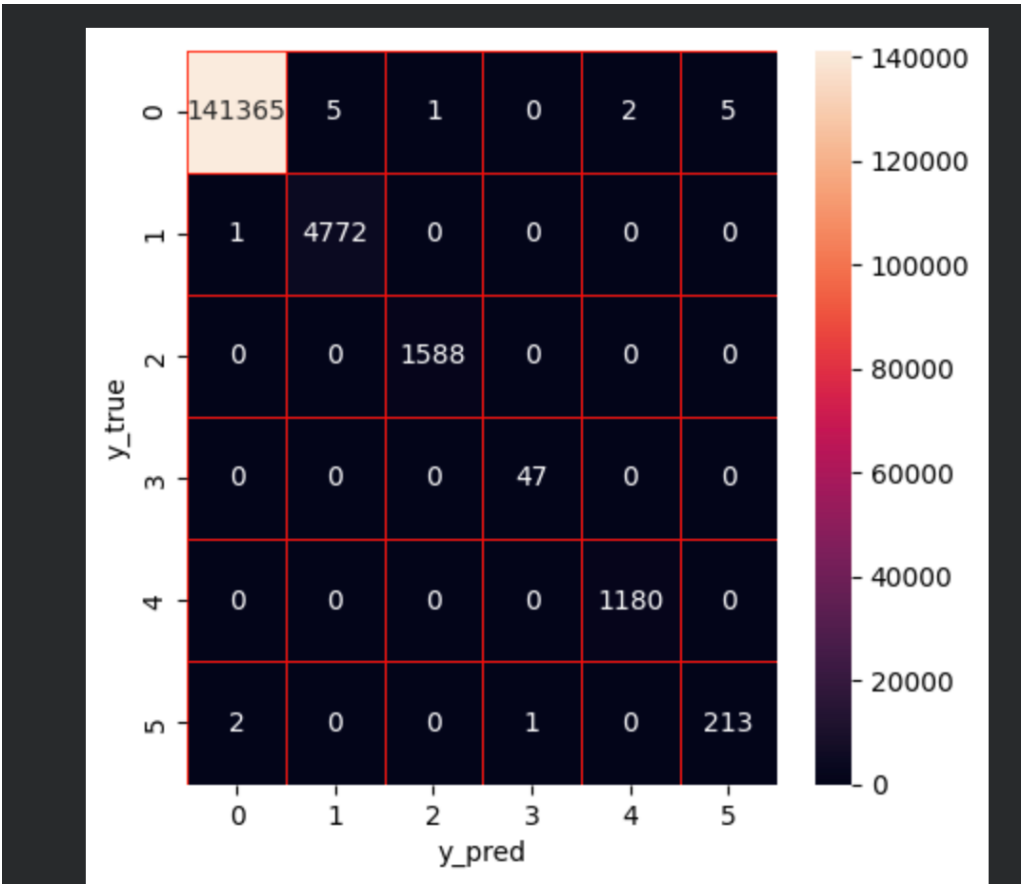


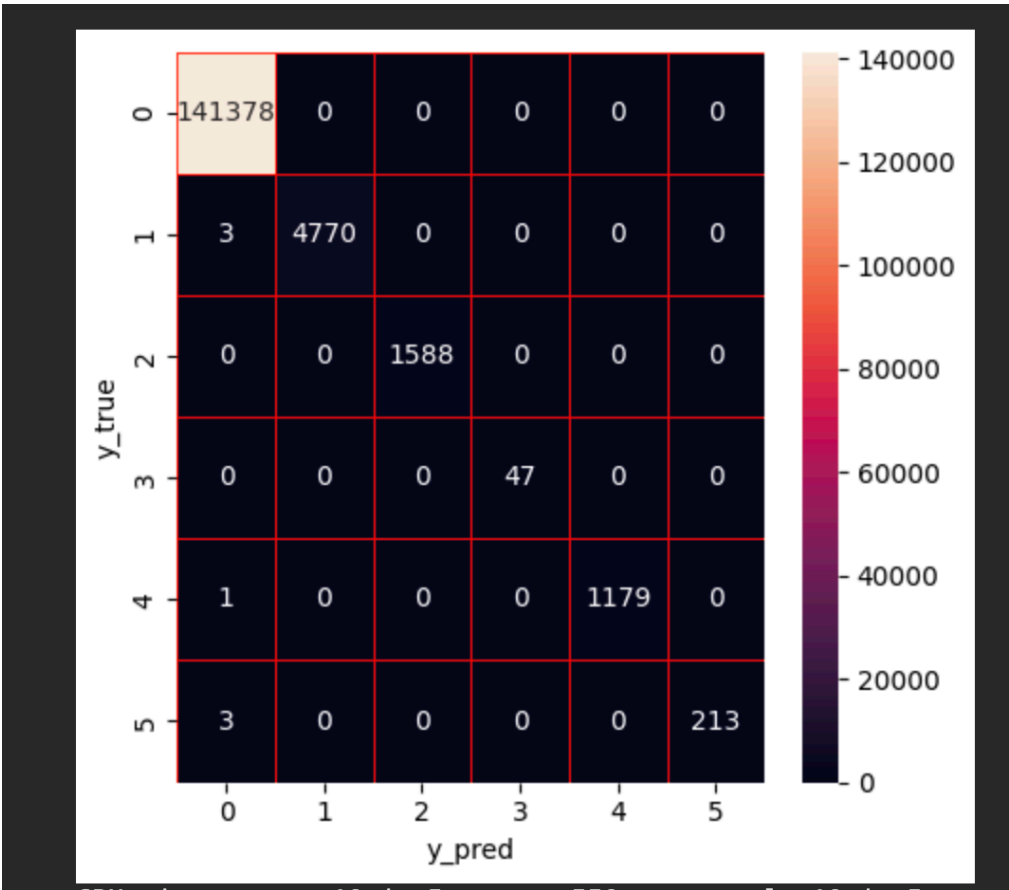
Decision Tree Results

Training time: 41.31432223320007					
Prediction time per sample: 0.00021021243155987867					
Cross-Validation scores: [0.99987431 0.99982404 0.99981901]					
Mean of Cross-Validation scores: 0.9998391224127615					
Accuracy of DT: 0.9998860452333391					
Precision of DT: 0.9998865459159304					
Recall of DT: 0.9998860452333391					
F1-score of DT: 0.9998862173322779					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	141378	
2	1.00	1.00	1.00	4773	
3	1.00	1.00	1.00	1588	
4	0.98	1.00	0.99	47	
5	1.00	1.00	1.00	1180	
6	0.98	0.99	0.98	216	
accuracy			1.00	149182	
macro avg	0.99	1.00	0.99	149182	
weighted avg	1.00	1.00	1.00	149182	



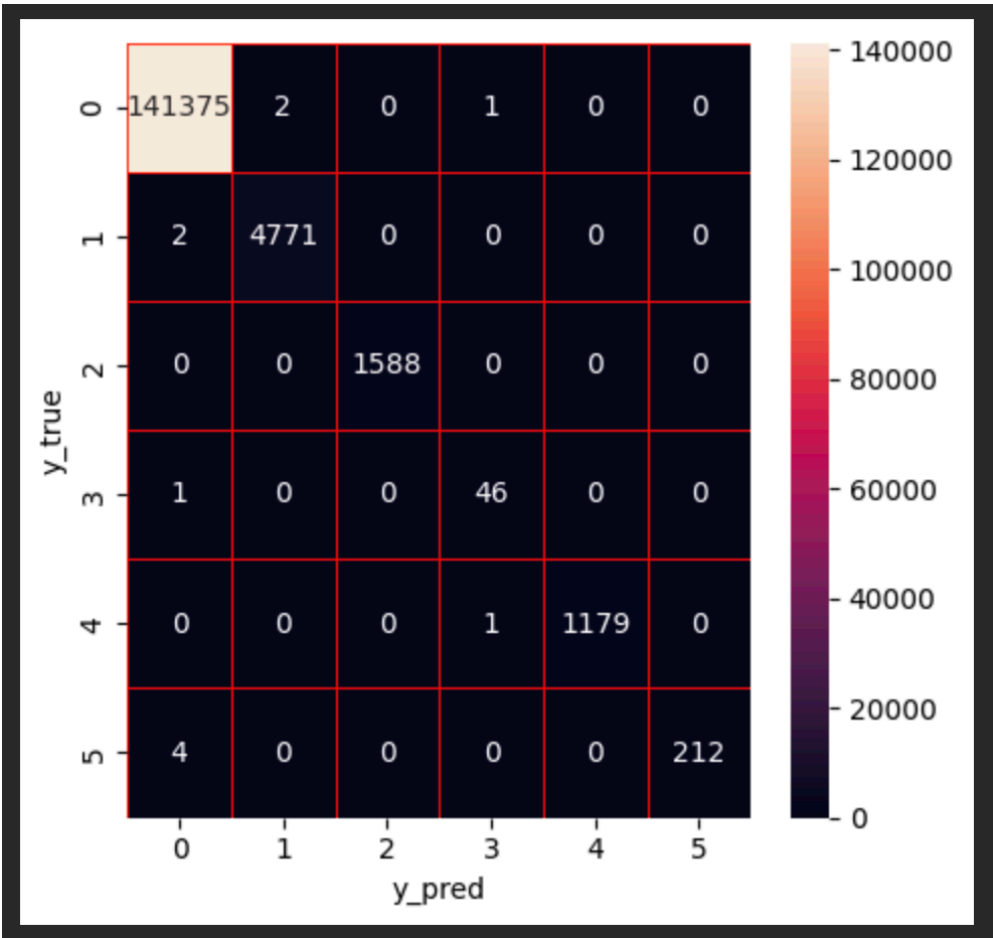
Random Forest Result

Training time: 206.1696331501007				
Prediction time: 0.00754397540417153 ms				
Cross-Validation scores: [0.99987934 0.99991453 0.99987431]				
Mean of Cross-Validation scores: 0.9998893966587735				
Accuracy of RF: 0.999953077449022				
Precision of RF: 0.999953079772167				
Recall of RF: 0.999953077449022				
F1-score of RF: 0.9999530031346308				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	141378
2	1.00	1.00	1.00	4773
3	1.00	1.00	1.00	1588
4	1.00	1.00	1.00	47
5	1.00	1.00	1.00	1180
6	1.00	0.99	0.99	216
accuracy			1.00	149182
macro avg	1.00	1.00	1.00	149182
weighted avg	1.00	1.00	1.00	149182



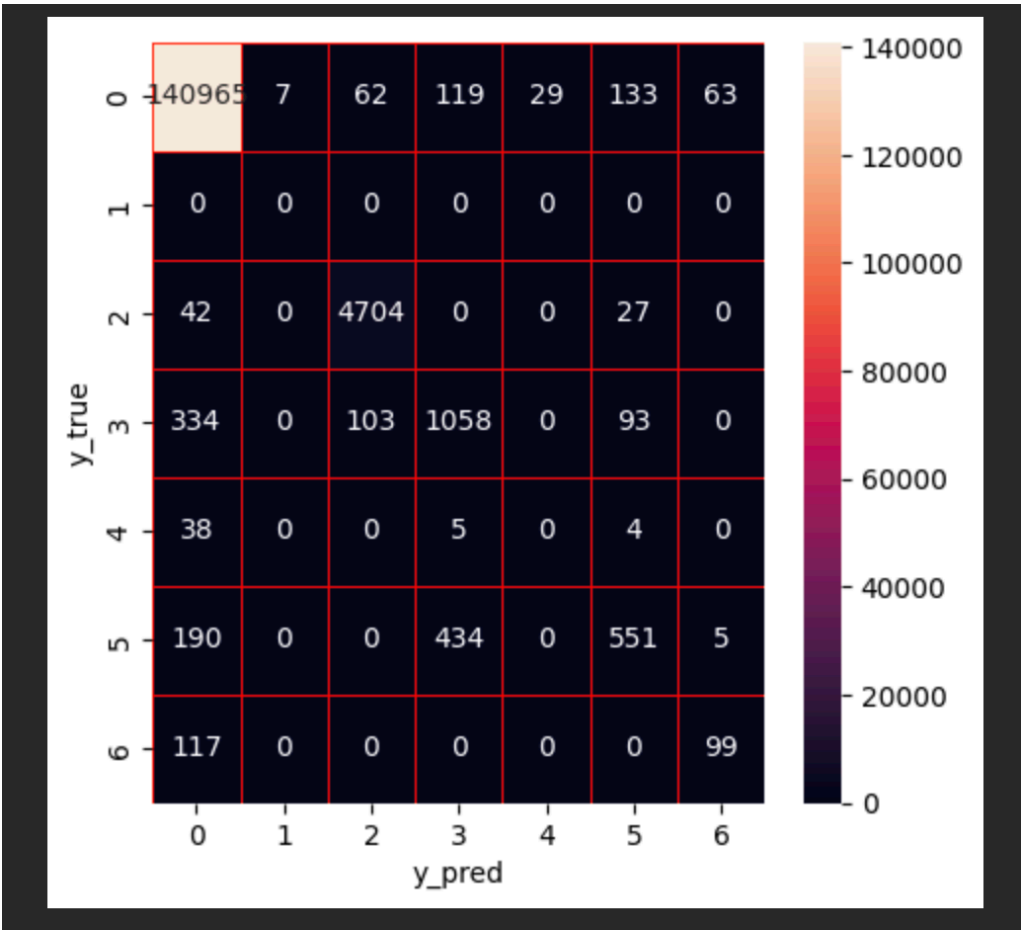
XGBoost Result

Training time: 84.21682858467102				
Prediction time per sample: 0.010709224884065524				
Cross-Validation scores: [0.99992962 0.99992459 0.99991453]				
Mean of Cross-Validation scores: 0.9999229128227816				
Accuracy of XGBoost: 0.9999262645627489				
Precision of XGBoost: 0.9999265451911871				
Recall of XGBoost: 0.9999262645627489				
F1-score of XGBoost: 0.9999262441625804				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	141378
2	1.00	1.00	1.00	4773
3	1.00	1.00	1.00	1588
4	0.96	0.98	0.97	47
5	1.00	1.00	1.00	1180
6	1.00	0.98	0.99	216
accuracy			1.00	149182
macro avg	0.99	0.99	0.99	149182
weighted avg	1.00	1.00	1.00	149182



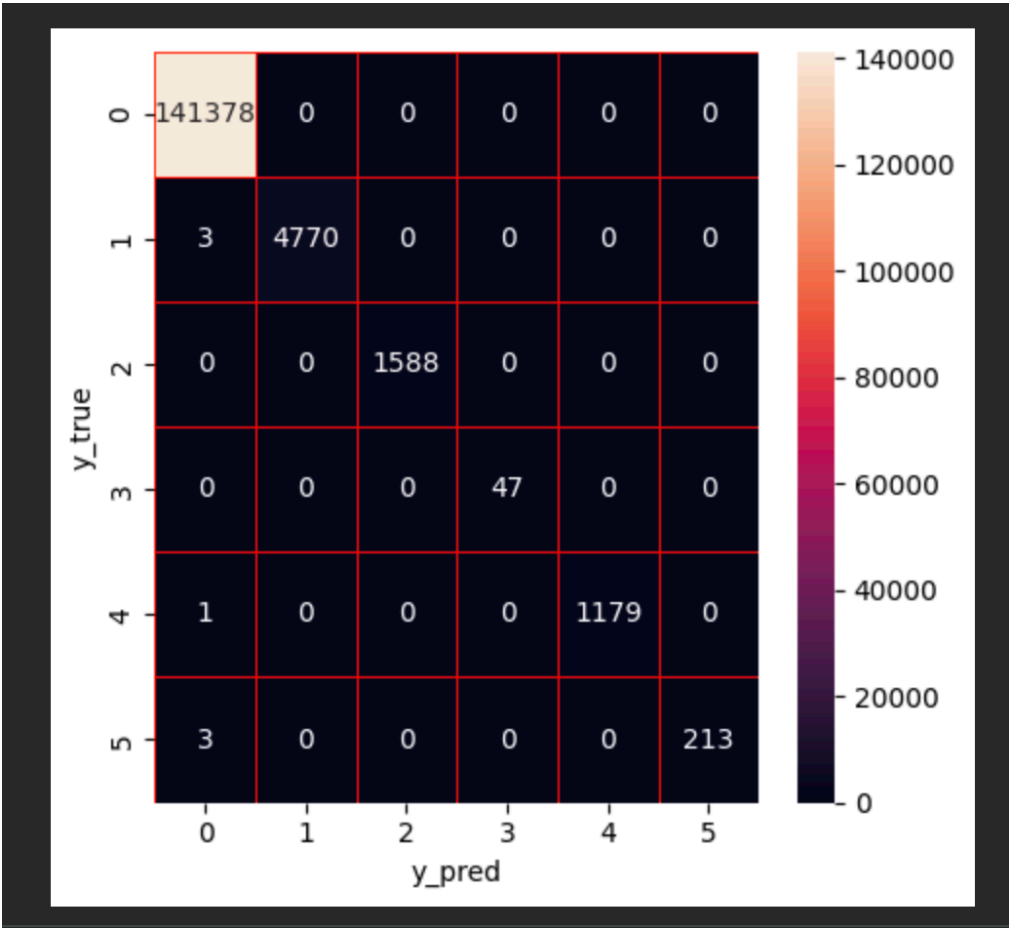
LightGBM Results

Accuracy of LightGBM: 0.9879006850692443				
Precision of LightGBM: 0.9869971913402779				
Recall of LightGBM: 0.9879006850692443				
F1-score of LightGBM: 0.9872726356658553				
	precision	recall	f1-score	support
0	0.99	1.00	1.00	141378
1	0.00	0.00	0.00	0
2	0.97	0.99	0.98	4773
3	0.65	0.67	0.66	1588
4	0.00	0.00	0.00	47
5	0.68	0.47	0.55	1180
6	0.59	0.46	0.52	216
accuracy			0.99	149182
macro avg	0.56	0.51	0.53	149182
weighted avg	0.99	0.99	0.99	149182



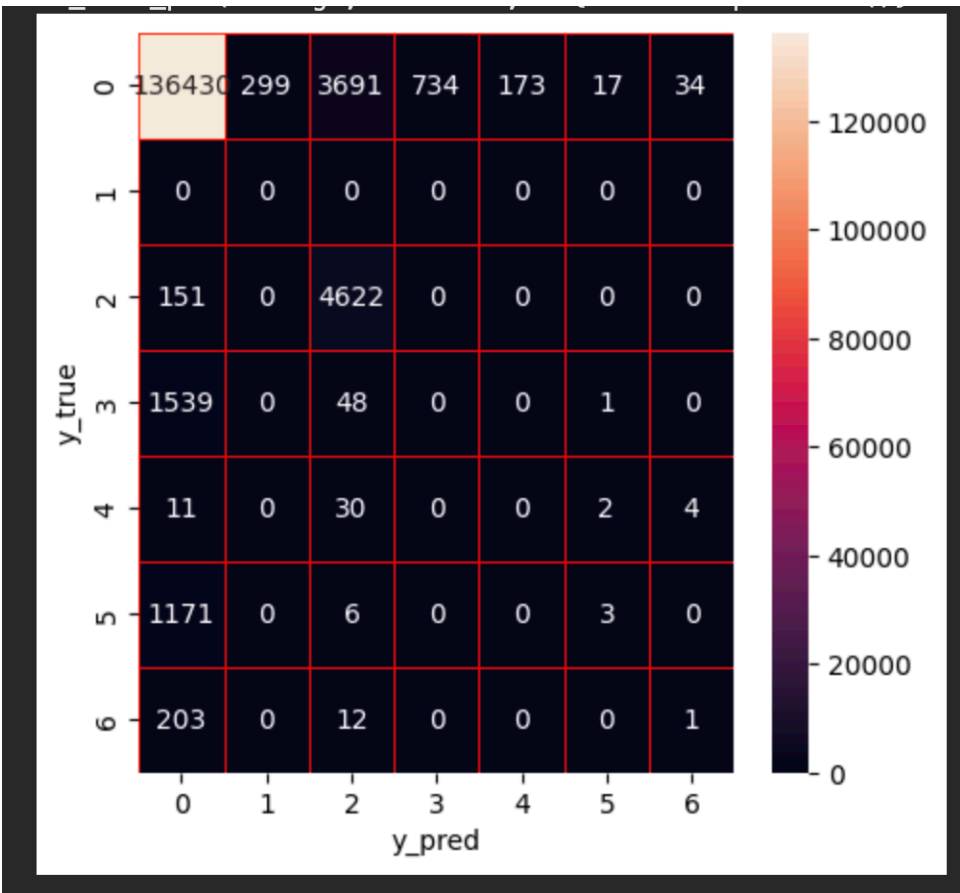
Model Selection_1 Results

Training time of Stacking: 18.649635553359985					
Testing time of Stacking: 0.003777386328999234					
Accuracy of Stacking: 0.999953077449022					
Precision of Stacking: 0.999953079772167					
Recall of Stacking: 0.999953077449022					
F1-score of Stacking: 0.9999530031346308					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	141378	
2	1.00	1.00	1.00	4773	
3	1.00	1.00	1.00	1588	
4	1.00	1.00	1.00	47	
5	1.00	1.00	1.00	1180	
6	1.00	0.99	0.99	216	
accuracy			1.00	149182	
macro avg	1.00	1.00	1.00	149182	
weighted avg	1.00	1.00	1.00	149182	



Model Selection_2 Results

Training time of Stacking: 31.431643962860107					
Testing time of Stacking: 0.008295234772946502					
Accuracy of Stacking: 0.9455296215361103					
Precision of Stacking: 0.9454534862619276					
Recall of Stacking: 0.9455296215361103					
F1-score of Stacking: 0.9431060067386635					
	precision	recall	f1-score	support	
0	0.98	0.97	0.97	141378	
1	0.00	0.00	0.00	0	
2	0.55	0.97	0.70	4773	
3	0.00	0.00	0.00	1588	
4	0.00	0.00	0.00	47	
5	0.13	0.00	0.00	1180	
6	0.03	0.00	0.01	216	
accuracy			0.95	149182	
macro avg	0.24	0.28	0.24	149182	
weighted avg	0.95	0.95	0.94	149182	



Automated Feature Selection Results

